

2(a)	to prevent the parent (fruit flies), reproducing / mating / AW, with the, first generation / offspring / AW ;	1
2(b)	<i>to separate male and female fruit flies:</i> use hand lens / (binocular) microscope / AW or ref. to difference between male and female (fruit flies) and <i>to place fruit flies:</i> use a, paint brush / pooter / forceps / AW ;	1
2(c)(i)	there is no difference between, the observed and expected, results ;	1

Question	Answer	Marks																								
2(c)(ii)	1 correct values of E ; 2 correct values of $(O-E)^2 / E$; <table><thead><tr><th>offspring phenotype</th><th>O</th><th>E</th><th>$(O-E)^2 / E$</th></tr></thead><tbody><tr><td>dark red eyes</td><td>691</td><td>702</td><td>0.172 121 / 702</td></tr><tr><td>brown eyes</td><td>260</td><td>234</td><td>2.889 676 / 234 = 26 / 9</td></tr><tr><td>bright red eyes</td><td>225</td><td>234</td><td>0.346 121 / 234</td></tr><tr><td>white eyes</td><td>72</td><td>78</td><td>0.462 36 / 78 = 6 / 13</td></tr><tr><td>total</td><td>1248</td><td>(1248)</td><td>3.869</td></tr></tbody></table> 3 $(\chi^2 =) 3.869$;	offspring phenotype	O	E	$(O-E)^2 / E$	dark red eyes	691	702	0.172 121 / 702	brown eyes	260	234	2.889 676 / 234 = 26 / 9	bright red eyes	225	234	0.346 121 / 234	white eyes	72	78	0.462 36 / 78 = 6 / 13	total	1248	(1248)	3.869	3
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2(c)(iii)	1 ref. to correct critical value = 7.815 ($p = 0.05$ and $df = 3$) ; 2 calculated value of, $\chi^2 / 3.869$, is less than 7.815 and null hypothesis is accepted ;	2																								
2(d)(i)	0.73 ;	1																								

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2(d)(ii)	any four from: <i>breeding experiment (MAX 3):</i> 1 ref. to the two genes (B / b and R / r), are not linked ; 2 <i>idea that</i> dominant alleles code for (functional) carrier, proteins / polypeptides ; 3 <i>idea that</i> in fruit flies with dark-red eyes entry of guanine and tryptophan (into organelle) allows pigments to be produced ; 4 fruit flies with white eyes, (have the genotype) bbrr / homozygous recessive (for both genes) or fruit flies with dark-red eyes are, B_R ; <i>chromatogram (MAX 3):</i> 5 fruit flies with dark-red eyes, have 4 pigments (in their eyes) ; 6 fruit flies with white eyes, have no pigments (in their eyes) ; 7 pigment 4 is drosopterin ; 8 <i>idea that</i>, ommochrome / brown pigment, is insoluble (in the chromatography solvent) or <i>idea that</i> none of the pigments in the chromatogram are, ommochrome / brown pigment ;	4